Power-board battery case comparison

Geometry, mounting assumptions and bare-board preload verification

8 November 2026 | Native structural analysis in ANSYS MAPDL 2026 R1.02

The mounting height determines whether the cells are already loaded

Placing the four tabs directly against the board creates approximately 1.18 mm of interference at the cell crowns. Each of the three cell solids shares about 1.70 cm^3 with the undeformed enclosure. A 3 mm tab stand-off clears the cells, nickel strips, washers and fasteners present in the 35-body mechanical STEP assembly. Thus a flush installation must displace or deform parts as the screws tighten. The raised case begins with an air gap. Translating a rigid cell along the PCB X axis first closes a gap after 0.146–0.325 mm, depending on the cell and direction. Flexible relative motion determines whether contact occurs under load.

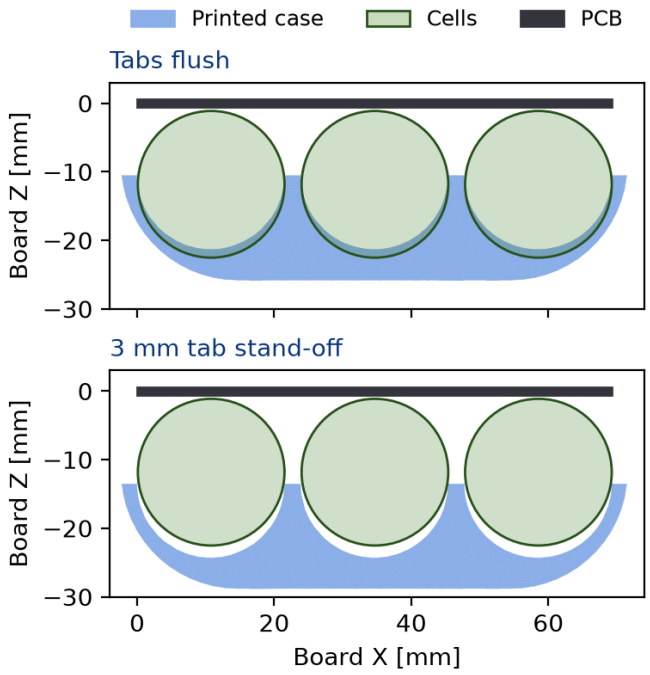


Figure 1: Section through the middle of the cells, in PCB coordinates. The solid outlines show as-designed geometry before tightening. The mounting tabs lie beyond this section.

Reference assembly

The October assembly has 208 bodies and a modeled mass of 251.82 g. Eight stand-off end faces carry the base motion. Two nickel strips attach each cell to the assembly. In the connected reference, six existing screws were loaded to 750 N apiece and locked. Its first mode was 485.964 Hz. Independent base excitation between 20 and 2000 Hz gave a maximum PCB directional displacement of 0.052674 mm RMS under Z excitation and a maximum PCB equivalent stress of 90.325 MPa RMS under X excitation. These archived quantities supply reference checks for the repeats.

Registration and fastener selection

The four nonplated corner holes in PCB1 ODB++ have finished diameter 4.200 mm. The hole pattern measures 62.865×114.427 mm. The case bores have two diameters, 5.000 and 5.090 mm, on a 62.855×113.241 mm pattern. Centring the two patterns leaves about 0.593 mm of offset at each end. The sum of the radial clearances of a nominal 4 mm shank in the smaller case bore and in the board is 0.600 mm. The approximately 0.007 mm difference is the available nominal alignment margin there. Printed hole size, fastener major diameter and drilling tolerances therefore matter for installation.

Tab stand-off [mm]	Cell overlap [mm^3/cell]	Washer bodies intersected	Other intersections
0	1696–1698	4	6 nickel strips
1.2715	0	4	None found
2	0	4	None found
3	0	0	None found

Table 1: Solid intersections between the unstrained case and mechanical STEP bodies. The board's lower face defines the tab seating datum.

Materials and mounting representation

The supplied single-solid STEP has volume 63.069 cm^3. At a printed-PLA reference density of 1.17 g/cm^3, a dense case weighs 73.79 g, about 29.3% of the bare assembly mass. The material axes place the print layers parallel to the PCB for this scenario. PolyLite PLA coupon tensile moduli are 3.427 GPa in the print plane and 3.065 GPa through the layers. Shear moduli and Poisson ratios are engineering assumptions. Actual infill, orientation, print parameters and case mass require measurement. The published coupons used 100% infill. Polymaker technical data sheet.

The candidate model uses four M4 beam shanks with an effective tensile area of 8.78 mm^2. The comparison uses 69 GPa for aluminium and an exploratory 3.0 GPa for nylon. The nominal nylon hardware is McMaster-Carr 93640A145, M4 $\times$ 0.7 thread, 12 mm long. The represented bearing patches transfer loads between each end and its solid mesh. Compression-only members represent tab seating, with an assumed PLA annulus at the raised attachment points. The four case screws have no measured installation torque. The 750 N in the reference applies to the six original battery fasteners. McMaster-Carr metric socket-head screws.

Numerical checks and remaining comparisons

The initial mesh has 16,108 curved quadratic tetrahedra. A 43,659-element refinement is available. Both have positive minimum scaled Jacobians. The native preparation check counted four screw beams, four compression seats and 8,038 candidate case contact faces. The contact calculation must establish residual penetration, support reactions, seated shape and stresses before modal or random-response comparisons can support a performance claim. In the October nickel-damage study an assumed material law first registered damage around 43 N during a partial pull. A physical breaking load remains undetermined.

The bare-board repeat reproduces the six locked bolt loads

The original battery attachment provides the initial stress field for the connected modal and vibration analyses. In the first nonlinear load step each of the six pretension elements receives 750 N. In the second step MAPDL holds the attained section adjustment fixed. The reaction at the pretension degree of freedom then measures how much load the locked joint retains. The four case screws are absent from this reference run.

Existing bolt	Locked adjustment [μm]	Locked reaction [N]
1	28.297	749.99935
2	22.205	749.99945
3	18.597	749.99908
4	30.457	749.99680
5	20.686	749.99942
6	18.772	749.99868

Table 2: Fresh MAPDL extraction after length locking. Every row agrees with the October export at its stored precision. The largest departure from the applied 750 N is 0.00321 N.

The solver stored eight equilibrium states and completed the two load steps in approximately 59 minutes on 12 distributed processes. The largest total displacement at lock was 0.0474595 mm. The repeated result differs from the October Mechanical export by 3.4×10^{-8} in relative terms. The model produced 45 warnings, including overlapping contacts, automatically removed redundant constraints and the fixed orientation of pretension elements during large-displacement analysis. The warning text printed in the main solver listing matched the reference run. This comparison establishes repeatability for the existing attachment. The effect of a case on preload, mode frequencies, cell motion and board stress requires the corresponding cased results.

How the side-pull comparison measures load transfer

The local model drives the outer cell sideways through its rigid pilot while the two nickel strips bend. In the washer-clamped fixture, each strip is compressed with 750 N and the clamp height is then locked. During the pull, the cell can translate in the other two directions and rotate. The cased fixture adds a deformable printed enclosure, four screw shanks and compression-only tab seats. The neighboring cells and screw tops are fixed in this local approximation. The native model check counted 9,502 potential PLA contact faces. A face transmits load only when its unilateral contact closes.

For each converged load increment the audit sums external forces and moments about the displaced cell centre. It includes the case screw supports and the neighboring-cell support when present. This makes it possible to test whether enclosure contact relieves the nickel strips at the cost of increased screw or board load. The check excluded unconverged diagnostic result sets. On historical friction fixtures it recovered the native force and moment sums across nominal, refined and reverse-direction pulls. Cased pull forces and damage thresholds await the loaded comparisons.